



**IT CLUB**

**Summer Internship 2025**

# ***IT CLUB SUMMER INTERNSHIP PROGRAM 2025***

**TASK - BASED LEARNING INTERNSHIP**



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## **IT Club Summer Internship 2025**

**Domain:** Machine Learning

**Duration:** 6 Weeks

### **Domain Overview:**

This document outlines the **Machine Learning** Internship Domain, a six-week program introducing interns to key concepts like supervised and unsupervised learning, model evaluation, and optimization. Through tasks and a final project, interns will apply ML techniques to solve practical problems and build a strong foundation.

### **What is Machine Learning?**

**Machine Learning (ML)** is a branch of artificial intelligence where computers learn from data to make decisions or predictions. Unlike traditional programming, ML allows systems to improve their performance with experience over time. In this internship, you'll explore foundational techniques in ML, focusing on data handling, model building, and evaluation, ultimately applying these skills in a real-world project.



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## Resources, Tools, and Setup Guide

### Tools & Platforms:

**1. Programming Language:** Python

**2. IDE:** Jupyter Notebook, Visual Studio Code or Google Colab

### **3. Libraries:**

- **NumPy** and **pandas** for data handling
- **scikit-learn** for ML algorithms
- **matplotlib** and **seaborn** for data visualization

**4. Version Control:** GitHub to manage and share code



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## Weekly Plan and Tasks

### Week 1: Introduction to Machine Learning

#### Tasks:

1. Set up the environment by installing **Python**, **Jupyter Notebook**, or **Google Colab**, and necessary libraries like **NumPy**, **pandas**, **scikit-learn**, **matplotlib**, and **seaborn**.
2. Load the **Iris** dataset (**or any similar dataset**) using **pandas** and explore the dataset by:
  - Viewing the first few rows (**head()** method).
  - Displaying the dataset's structure (columns, types, null values, etc.).
3. Visualize the dataset with basic plots:
  - Create a scatter plot of any two features.
  - Plot the distribution of a feature (e.g., sepal length).



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## Weekly Plan and Tasks

### Week 2: Data Preprocessing and Feature Engineering

#### Tasks:

**1. Handle missing values by using **pandas**:**

- Identify missing values in a dataset.
- Use imputation methods (mean, median, or mode) to fill them.

**2. Perform feature scaling:**

- Normalize the dataset using **Min-Max Scaling**.
- Standardize features using **z-score scaling**.

**3. Encode categorical data:**

- Apply **One-Hot Encoding** on a categorical column.
- Use **Label Encoding** for another categorical column.

**4. Perform exploratory data analysis (EDA):**

- Plot correlations using a **heatmap**.
- Visualize relationships using **pair plots**.



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## Weekly Plan and Tasks

### Week 3: Regression Algorithms

#### Tasks:

1. Train a Linear Regression model on a real dataset such as **Boston Housing** (e.g., predicting housing prices).
2. Evaluate the model's performance using the following metrics:
  - **Mean Absolute Error (MAE)**
  - **Mean Squared Error (MSE)**
  - **R-squared score**
3. Implement **Polynomial Regression** for non-linear data



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## Weekly Plan and Tasks

### Week 4: Classification Algorithms

#### Tasks:

1. Train a **Logistic Regression classifier** on a dataset (e.g., Iris dataset for binary classification).
2. Evaluate the model's performance using the following metrics:
  - **Mean Absolute Error (MAE)**
  - **Mean Squared Error (MSE)**
  - **R-squared score**
3. Explore additional classifiers:
  - Train a **Decision Tree classifier**.
  - Implement a **K-Nearest Neighbors (KNN)** model.
4. Create confusion matrices to visualize predictions for each classifier.



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## Weekly Plan and Tasks

### Week 5: Unsupervised Learning and Clustering

#### Tasks:

**1. Use K-Means clustering:**

- Cluster a dataset into 3 groups and visualize the results.
- Plot the **elbow method** to find the optimal number of clusters.

**2. Apply Principal Component Analysis (PCA) for dimensionality reduction:**

- Reduce the dataset to 2 or 3 components.
- Visualize the transformed dataset using a scatter plot.



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## **Weekly Plan and Tasks**

### **Week 6: Real-World Project - Predictive Model Development**

#### **Project Title: Predicting Customer Churn with Machine Learning**

**Objective:** Build a complete predictive model for identifying customers at risk of churn in a business setting.

#### **Steps:**

- 1. Data Preprocessing:** Clean and preprocess the dataset, handling missing values and performing feature engineering.
- 2. Feature Engineering:** Encode categorical data and select the most important features.
- 3. Model Selection and Training:** Train a classification model (e.g., Random Forest or XGBoost) to predict churn.
- 4. Hyperparameter Tuning:** Experiment with hyperparameters to optimize model performance.
- 5. Evaluation:** Evaluate the model with metrics such as accuracy, F1-score, and ROC-AUC.
- 6. Documentation:** Summarize your approach, methods, and findings in a report.



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## **Deliverables:**

- **Code Files:** Submit the complete code in Jupyter Notebook or .py files.
- **Project Report:** 2-3 pages outlining the project's objectives, methodology, and results.
- **GitHub Repository:** Host your project on GitHub and include the link.



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## **Submission Guidelines**

- **Weekly Tasks:** Submit each week's tasks via Google Forms, attaching code files and GitHub repository links where applicable.
- **GitHub Repository:** Create a repository for each week's code and final project files.
- **LinkedIn Showcase:** At the end of each week, post a brief reflection on LinkedIn about the tasks completed and key learnings.  
(Record task demonstrations using any recorder like OBS Studio.)
- **Final Submission:** For the final project, submit code, GitHub repository link, and LinkedIn post link via Google Forms.

**Submission Form Link for Weekly Tasks:**

[\*\*ML Internship Tasks Submission\*\*](#) (Clickable Link)

**Best of Luck!**